

```

#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>
#include <time.h>

int main()
{
    int fd[2];
    char msg[] = "Hello from Parent";
    char buffer[100];

    pipe(fd);

    clock_t start = clock();

    int pid = fork();

    if (pid == 0)
    {
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Child consumed: %s\n", buffer);

        close(fd[0]);
    }
    else
    {
        close(fd[0]);

        printf("Parent produced: %s\n", msg);

        write(fd[1], msg, strlen(msg) + 1);

        close(fd[1]);
    }
}

```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-I To Bracket
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-G Copy ^B Where Was

```

varshith@VarhithReddy:~$ ./q5a
Parent produced: Hello from Parent
Child consumed: Hello from Parent
Communication time: 0.000162 seconds

```

```
varshith@VarhithReddy: ~  
GNU nano 8.7.1 q5b.c  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    int fd[2];  
    pipe(fd);  
  
    int pid1 = fork();  
    if (pid1 == 0)  
    {  
        dup2(fd[1], STDOUT_FILENO);  
        close(fd[0]);  
        close(fd[1]);  
        execlp("ls", "ls", "-l", NULL);  
    }  
  
    int pid2 = fork();  
    if (pid2 == 0)  
    {  
        dup2(fd[0], STDIN_FILENO);  
        close(fd[1]);  
        close(fd[0]);  
        execlp("grep", "grep", ".c", NULL);  
    }  
  
    close(fd[0]);  
    close(fd[1]);  
}
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was

```
varshith@VarhithReddy: ~$ gcc q5b.c -o q5b  
varshith@VarhithReddy: ~$ ./q5b  
-rw-r--r-- 1 varshith varshith 88 Jul 28 04:46 addition.c  
-rw-r--r-- 1 varshith varshith 290 Jul 28 05:45 arithmetic.c  
-rw-r--r-- 1 varshith varshith 231 Aug 25 05:49 nano2.c  
-rw-r--r-- 1 varshith varshith 551 Aug 1 10:12 process.c  
-rw-r--r-- 1 varshith varshith 777 Sep 3 04:51 q5a.c  
-rw-r--r-- 1 varshith varshith 551 Sep 3 04:52 q5b.c  
-rw-r--r-- 1 varshith varshith 207 Aug 3 07:01 sample.c  
-rw-r--r-- 1 varshith varshith 117 Aug 3 07:08 sample1.c  
-rw-r--r-- 1 varshith varshith 213 Aug 3 08:06 sample2.c  
-rw-r--r-- 1 varshith varshith 167 Aug 3 08:31 sample3.c  
-rw-r--r-- 1 varshith varshith 1129 Jul 28 05:57 switch.c  
-rw-r--r-- 1 varshith varshith 563 Aug 11 05:23 week1.c
```